



Generative AI Training

Gen AI use-cases
across job functions and industries

**The economic potential of
generative AI: The next
productivity frontier**

Goldman Sachs CEO David Solomon:

“AI can now draft **95%** of an S1 IPO prospectus **in minutes** (a job that used to require a 6-person team multiple weeks). The last 5% now matters because the rest is now a **commodity**.”

The AI effect
at its finest.



The background is a dark blue gradient. On the left, there are two horizontal strings of binary code. The top string is cyan and the bottom string is magenta. From the right end of each string, a fan of lines radiates outwards. The cyan lines fan out to the right, and the magenta lines fan out to the right, overlapping with the cyan lines. The lines are thin and numerous, creating a dense, web-like pattern on the right side of the image. The lines are colored in shades of cyan, magenta, and yellow.

101001000001101101000

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Where business value lies

<https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/the-economic-potential-of-generative-ai-the-next-productivity-frontier#introduction>

Customer Operations

**Customer operations:
Improving customer and agent experiences**

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Potential productivity lift
Customer operations

38%
of global functional
spending

~\$404B

Generative AI has the potential to revolutionize the entire customer operations function, improving the customer experience and agent productivity through digital self-service and enhancing and augmenting agent skills. The technology has already gained traction in customer service because of its ability to automate interactions with customers using natural language. Research found that at one company with 5,000 customer service agents, the application of generative AI increased issue resolution by 14 percent an hour and reduced the

Source: McKinsey Digital

Customer Operations / Support use-cases

- Intelligent support agent
- Voice assisted
- Trained on company policies, tone, approach
- Agent will have access to real time product and purchase information

Source: McKinsey Digital

Customer Operations / Support use-cases

Research found that at one company with 5,000 customer service agents, the application of generative AI :

- Increased issue resolution by 14 percent an hour
- Reduced the time spent handling an issue by 9 percent.
- Reduced agent attrition and requests to speak to a manager by 25 percent.
- Productivity and quality of service improved most among less-experienced agents

Marketing and Sales

Marketing and sales: Boosting personalization, content creation, and sales productivity

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Potential productivity lift
Marketing

10%

of global functional
spending

~\$463B

Sales

4%

of global functional
spending

~\$486B

Source: McKinsey Digital

Marketing and Sales use-cases

- Marketing communications
 - Targeted
 - Personalized
 - Brand content creation (images, articles, blogs, social media posts)
 - SEO optimization
- Sales lead generation and lead development
 - Market research (segments, geographies, demographics)
 - Lead research
- Lead nurturing
- Competitive analysis and positioning
- SWOT Analysis

Marketing and Sales use-cases

Marketing demo:

Hyundai - AI powered Instagram Creativity

<https://www.hyundai.news/eu/articles/press-releases/santa-fe-ai-generated-campaign-invites-instagram-users-to-dream-destinations.html>

Product research demo

Lead nurturing demo

Software Engineering

**Software engineering:
Speeding
developer work as
a coding assistant**

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Potential productivity lift
Software engineering in corporate IT

31%

of global functional
spending

~\$485B

Software engineering in product development

32%

of global functional
spending

~\$414B



Generative AI impacts the full Software Development Lifecycle



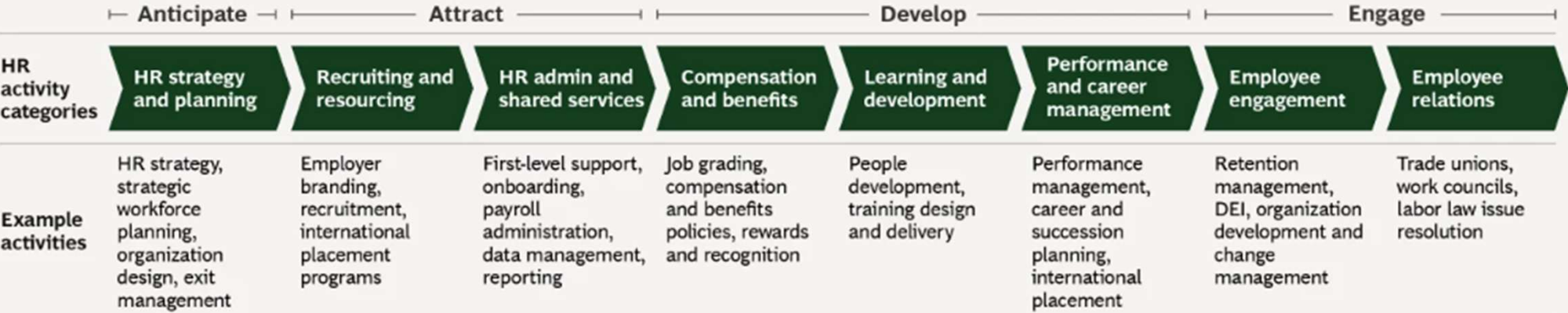
https://azuremarketplace.microsoft.com/en-us/marketplace/consulting-services/capgemini-group.gen_ai_software_engineering_github_copilot

Software Engineering use-cases

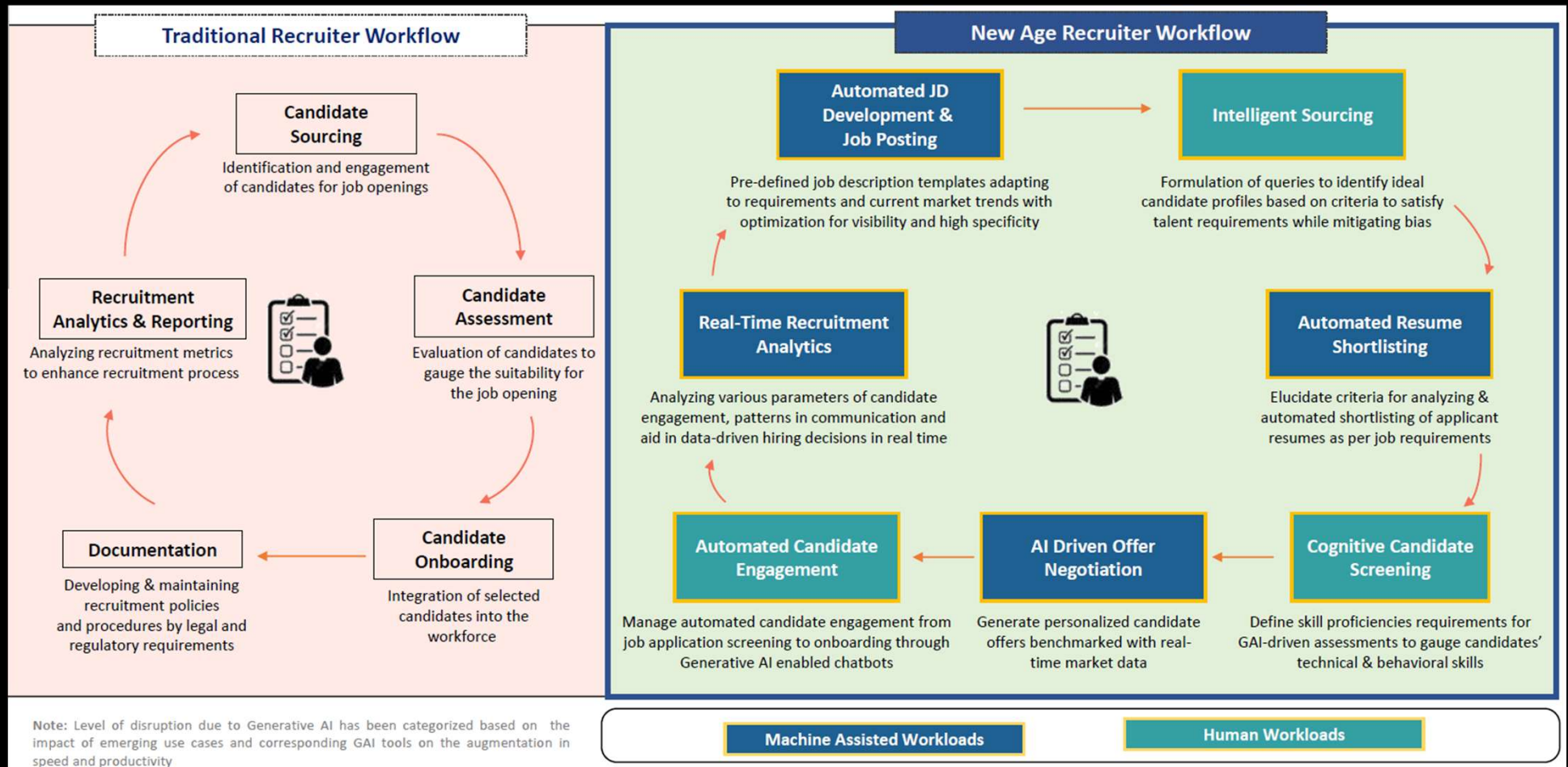
- Requirements analysis
- Technical architecture and choices generation keeping platform, cost, performance requirements in mind
- Frontend / UI Code generation from mockups
- Development (database, schema, middle tier code) with Gen AI
- Test case generation, test code generation, tests execution, tests results analysis
- Test failure analysis and bugfixing
- Self healing!!

Human Resources (HR), Learning & Development (L&D)

HR transformation | GenAI has the potential to drive about 30% increased productivity across the HR value chain in the near term



Talent acquisition transformation



Gen AI in Finance functions (CFO)

Workflow Steps With AI/GenAI Applications	Financial Data Collection and Consolidation	Data Analysis and Interpretation	Report Generation	Distribution and Dissemination
	Data Gathering	Data Cleaning and Transformation	Drafting Reports	Report Distribution
	Data Standardization	Budget and Forecast Comparison	Review and Approval	Access Management
	Data Validation	KPI Calculation and Monitoring		Feedback Collection
	Data Aggregation	Quality Checks and Assurance		
Target Personas	Financial Analyst	Financial Analyst	Financial Analyst	Financial Analyst
	Controller	Controller	Controller	Controller
	Auditor	Auditor	CFO	CFO
	Compliance Officer	CFO Compliance Officer	Auditor	Compliance Officer

<https://www.leewayhertz.com/ai-for-financial-reporting/#benefits-of-ai-in-financial-reporting>

Product Research and Development



**Product R&D:
Reducing
research and
design time,
improving
simulation and
testing**

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Potential productivity lift
Product R&D

12%

of global functional
spending

~\$328B

Generative AI's potential in R&D is perhaps less well recognized than its potential in other business functions. Still, our research indicates the technology could deliver productivity with a value ranging from 10 to 15 percent of overall R&D costs.

For example, the life sciences and chemical industries have begun using generative AI foundation models in their R&D for what is known as generative design. Foundation models can generate candidate molecules, accelerating the process of developing new drugs and materials. Entos, a biotech pharmaceutical company, has paired generative AI with automated synthetic

Product Research and Development

Boeing is using Generative AI to design airplane parts

BMW – Digital Twin virtual factory

World's First Virtual Factory

BMW Group Opens Debrecen
Factory in NVIDIA Omniverse



<https://www.youtube.com/watch?v=g78YHYXXils>

Toyota – Generative designs for vehicles



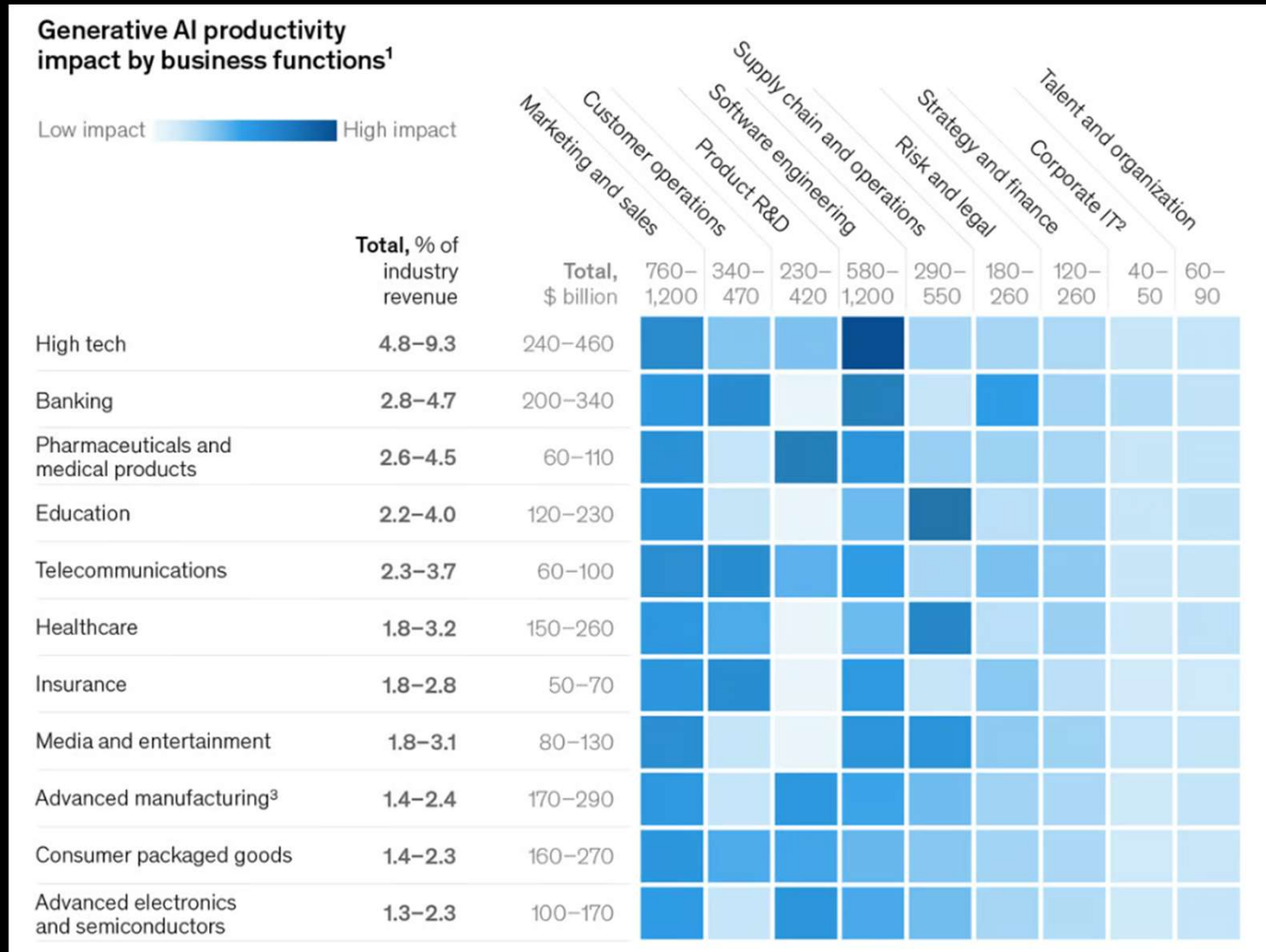
Generative AI Technique for Vehicle Design

<https://www.youtube.com/watch?v=xkwwIDgrmbk>



Industry impacts

Impact across industries



	Total value potential per industry, \$ billion (% of industry revenue)	Value potential, as % of operating profits ¹	Product R&D, software engineering	Customer operations	Marketing and sales	Other functions
Banking	200–340 (3–5%)	9–15	Legacy code conversion Optimize migration of legacy frameworks with natural-language translation capabilities	Customer emergency interactive voice response (IVR) Partially automate, accelerate, and enhance resolution rate of customer emergencies through generative AI-enhanced IVR interactions (eg, for credit card losses)	Custom retail banking offers Push personalized marketing and sales content tailored for each client of the bank based on profile and history (eg, personalized nudges), and generate alternatives for A/B testing	Risk model documentation Create model documentation, and scan for missing documentation and relevant regulatory updates
Retail and consumer packaged goods²	400–660 (1–2%)	27–44	Consumer research Accelerate consumer research by testing scenarios, and enhance customer targeting by creating “synthetic customers” to practice with	Augmented reality–assisted customer support Rapidly inform the workforce in real time about the status of products and consumer preferences	Assist copy writing for marketing content creation Accelerate writing of copy for marketing content and advertising scripts	Procurement suppliers process enhancement Draft playbooks for negotiating with suppliers
Pharma and medical products	60–110 (3–5%)	15–25	Research and drug discovery Accelerate the selection of proteins and molecules best suited as candidates for new drug	Customer documentation generation Draft medication instructions and risk notices for drug resale	Generate content for commercial representatives Prepare scripts for interactions with physicians	Contract generation Draft legal documents incorporating specific regulatory

Optimizing field operations with AI-ML



https://www.youtube.com/watch?v=znz5_VbzxPU

Automotive – in vehicle experience



The Mercedes-Benz Intelligent Vehicle Assistant,

"Hey Mercedes," offers a hyper-personalized user experience

Leveraging large-scale Generative AI models and Azure OpenAI Services GPT-4 Turbo with vision, it integrates seamlessly with multimodal components to provide real-time assistance and contextual information.

<https://www.youtube.com/watch?v=ocxnhqZuS8w>

Moody's Copilot



<https://www.youtube.com/watch?v=7O8jtu3bpS8>

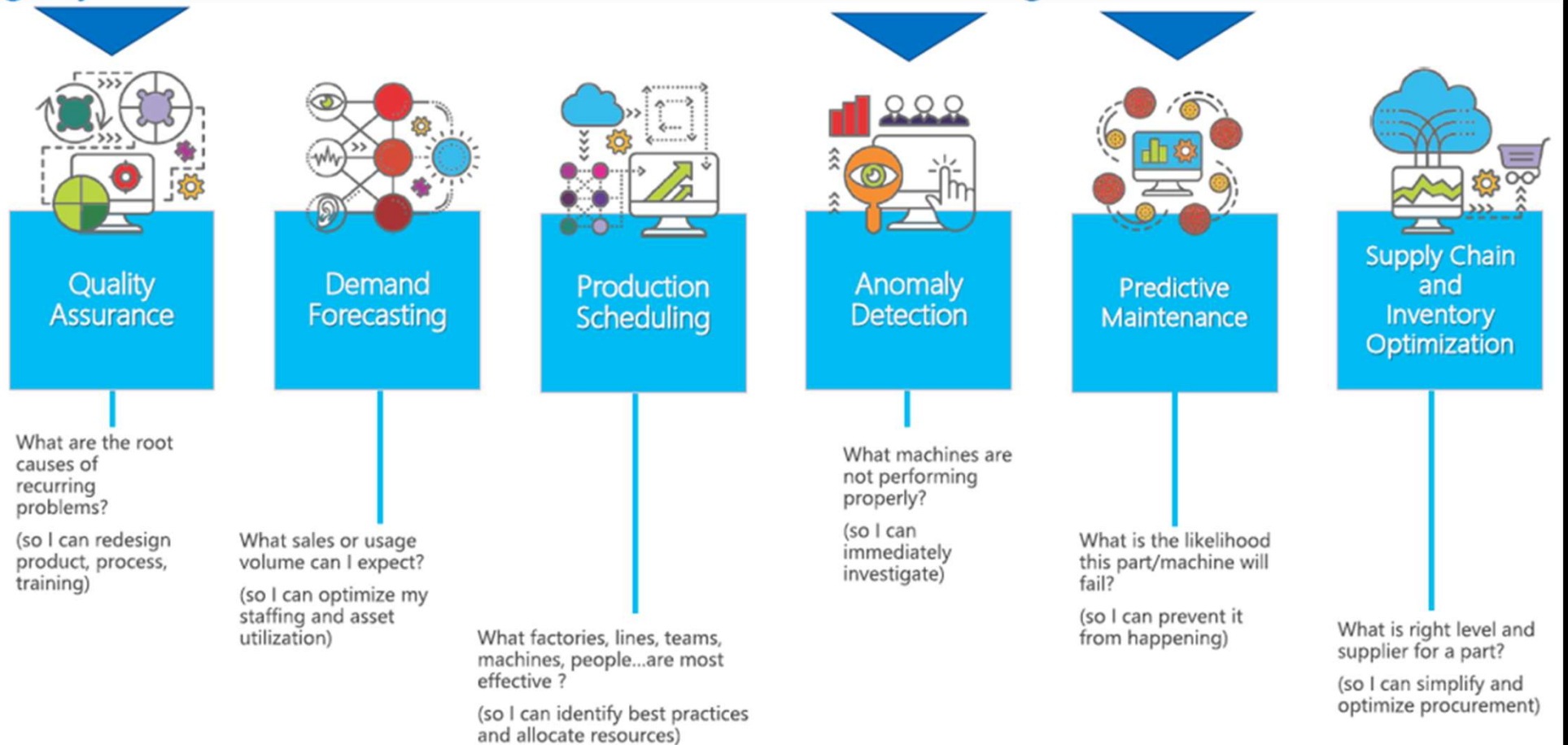
Chemical plant automation with AI



JSR Chemical Plant

<https://www.youtube.com/watch?v=g4H6Fkag3cY>

High yield use-cases in manufacturing

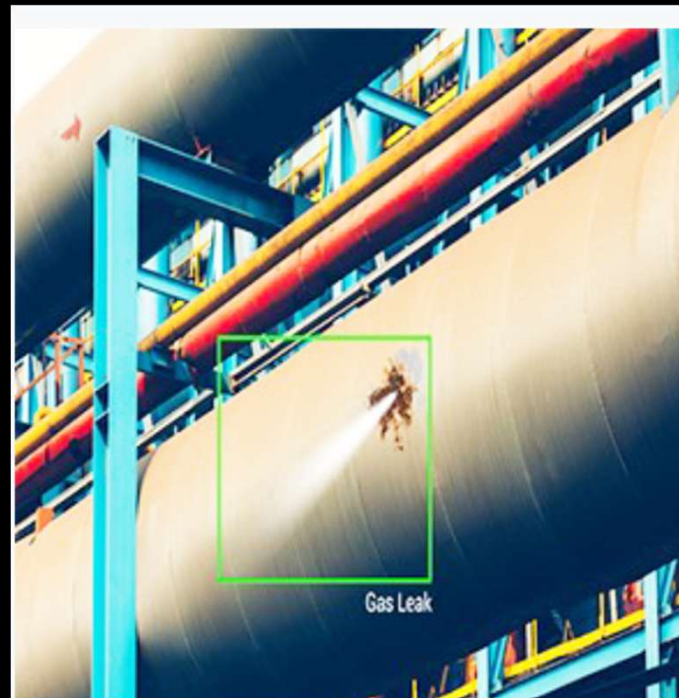


Predictive maintenance

Predictive maintenance - vision model approach

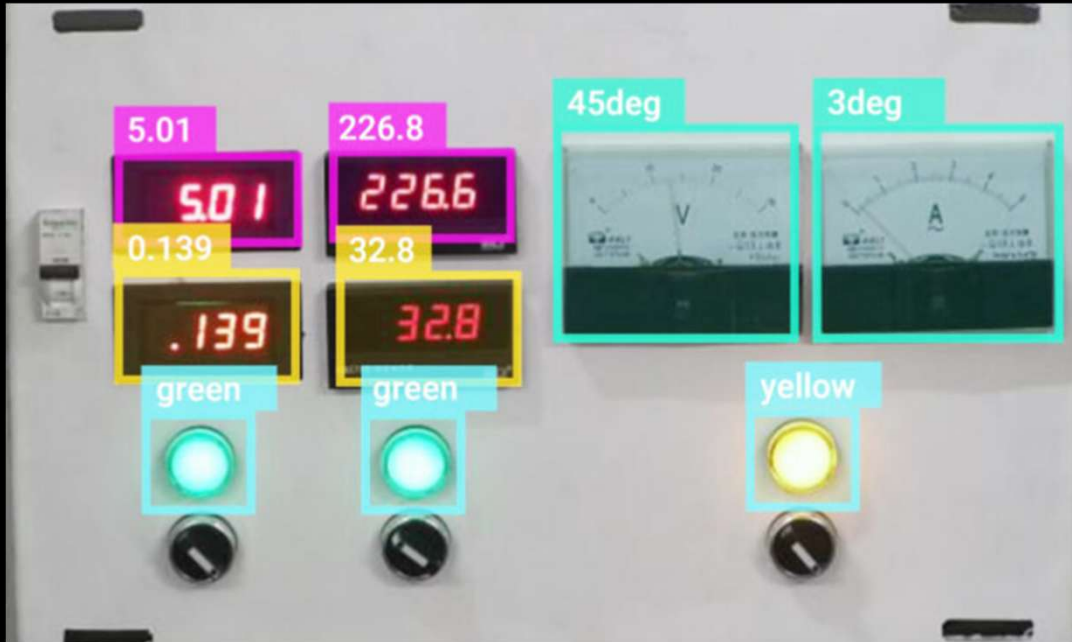


Corrosion detection



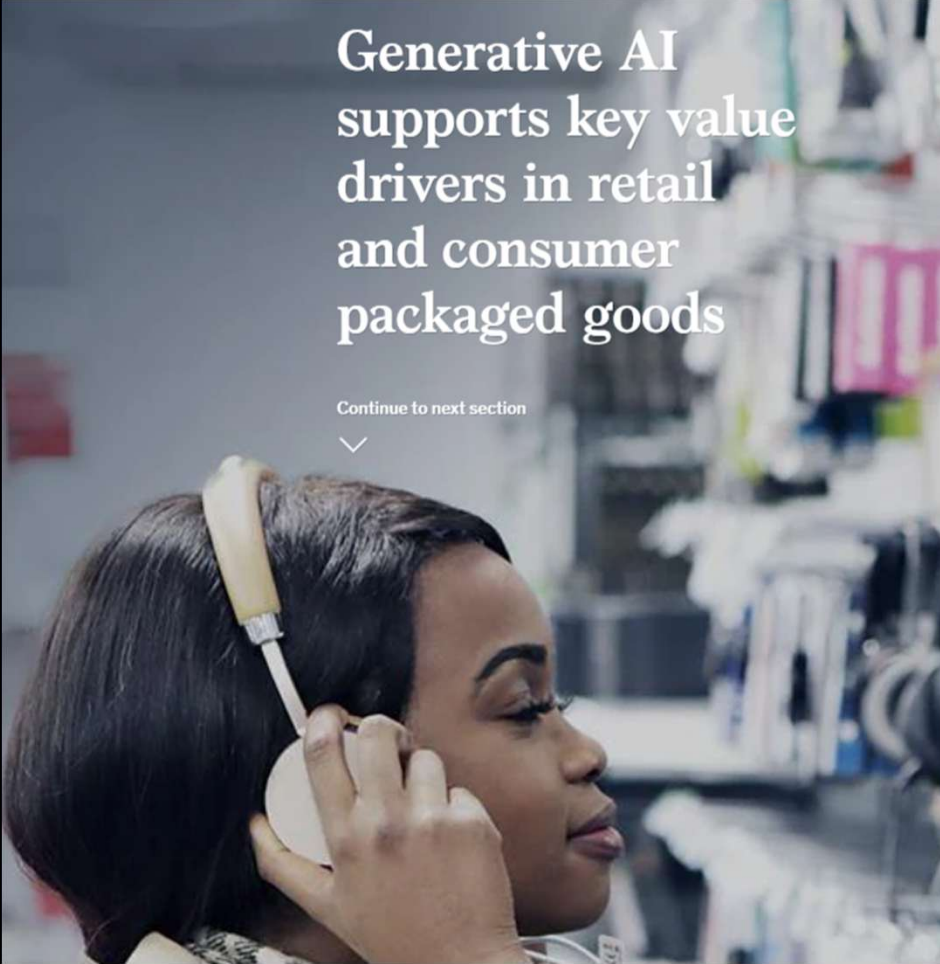
Leak detection

Predictive maintenance



Remote monitoring and automated event notifications by reading analogue gauges with computer vision

Retail



Generative AI
supports key value
drivers in retail
and consumer
packaged goods

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Potential productivity lift
Retail and consumer packaged goods

1-2%
of global industry
revenue

**~\$400B–
\$660B**

The technology could generate value for the retail and consumer packaged goods (CPG) industry by increasing productivity by 1.2 to 2.0 percent of annual revenues, or an additional \$400 billion to \$660 billion.^[1] To streamline processes, generative AI could automate key functions such as customer service, marketing and sales, and inventory and supply chain management. Technology has played an essential role in the retail and CPG industries for decades. Traditional AI and advanced analytics solutions have helped companies manage vast pools of data across large numbers of SKUs, expansive supply chain and warehousing networks, and complex product categories such as consumables. In addition, the industries are heavily customer facing, which offers opportunities for generative AI to complement previously existing artificial intelligence. For example, generative AI's ability to personalize offerings could optimize marketing and sales activities already handled by existing AI solutions. Similarly, generative AI tools excel at data management and could support existing AI-driven pricing tools. Applying

Retail example: Stitchfix

Stitch Fix, which uses algorithms to suggest style choices to its customers, has experimented with DALL·E to visualize products based on customer preferences regarding color, fabric, and style.

Using text-to-image generation, the company's stylists can visualize an article of clothing based on a consumer's preferences and then identify a similar article among Stitch Fix's inventory.

Banking

Why banks could realize significant value

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Potential productivity lift
Banking

3-5%
of global industry
revenue

**~\$200B–
\$340B**

Generative AI could have a significant impact on the banking industry, generating value from increased productivity of 2.8 to 4.7 percent of the industry's annual revenues, or an additional \$200 billion to \$340 billion. On top of that impact, the use of generative AI tools could also enhance customer satisfaction, improve decision making and employee experience, and decrease risks through better monitoring of fraud and risk.

Banking, a knowledge and technology-enabled industry, has already benefited significantly from previously existing applications of artificial intelligence in areas such as marketing and customer operations.^[1] Generative AI applications could deliver additional benefits, especially because text modalities are prevalent in areas such as regulations and programming language, and the industry is customer facing, with many B2C and small-business customers.^[2]

Pharmaceuticals

A woman in a lab coat is looking through a microscope. The background is a soft blue gradient.

Pharmaceuticals
and medical
products could see
benefits across the
entire value chain

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3-5%
of global industry
revenue

**~\$60B–
\$110B**

Our analysis finds that generative AI could have a significant impact on the pharmaceutical and medical-product industries—from 2.6 to 4.5 percent of annual revenues across the pharmaceutical and medical-product industries, or \$60 billion to \$110 billion annually. This big potential reflects the resource-intensive process of discovering new drug compounds. Pharma companies typically spend approximately 20 percent of revenues on R&D,^[1] and the development of a new drug takes an average of ten to 15 years. With this level of spending and timeline, improving the speed and quality of R&D can generate substantial value. For example, lead identification—a step in the drug discovery process in which researchers identify a molecule that would best address the target for a potential new drug—can take several months even with “traditional” deep learning techniques. Foundation models and generative AI can enable organizations to complete this step in a matter of weeks.

Media and Entertainment



<https://www.youtube.com/watch?v=q2ITXhM-53I>